Health defects of PM 2.5 mapped

Marnix Kuil 2900391, Peter Ac 2877986 Simon Lamers 2894404,Sophie Valckx 2905390, Thijmen Gerrits

2026-06-23

Set-up your environment

```r
require(tidyverse)
```

```
## Loading required package: tidyverse

## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.1     v readr     2.2.0
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.3     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.2
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
require(rmarkdown)
```

```
## Loading required package: rmarkdown
```

```r
require(yaml)
```

```
## Loading required package: yaml
```

Title Page

Marnix Kuil, Peter Ac, Simon Lamers, Sophie Valckx, Thijmen Gerritsen, Thomas Smit

Tutorial group number 3.5

Tutorial lecturer Chantal Schouwenaar

Part 1 - Identify a Social Problem

1.1 Describe the Social Problem

The problem that we will assess is particulate matter distribution across different regions in the Netherlands, and its correlation with health effects, more specifically, mortality rates. The datasets that we use measure PM2.5, this is particulate matter that is less than 2.5 m in size. Another frequently used measurement of particulate matter is PM10, which is particulate matter that is less than 10 m in size. We have chosen to use PM2.5 because its small size allows it to penetrate deeper into the lungs than larger particles (WHO, 2006).

A potential pitfall in our research is that Janssen et al. (2013) argues that both PM2.5 and PM10 have a very similar association with mortality. So when we focus on PM2.5, we should not discount the fact that PM10 is also heavily correlated with public health issues. However, Janssen et al. (2013) draws on data from 2008/2009, being aware of outdated measurements is of importance in our research.

Another gap in the research is the lack of structural measurement of ultrafine particles (UFP) in the Netherlands. These particles are smaller than PM2.5 and are also commonly associated with health risks. Some of the health risks that are commonly attributable to PM2.5 may partially have a causal relationship with UFP concentration, which is much more difficult to find consistent data for (CLO, 2024).

This is a relevant issue in Dutch society for a variety of reasons. For example, the debate around the role that TATA Steel plays when it comes to contributing to health risks for local citizens. For example, residents in Wijk aan Zee lose an average of 2.5 months of life expectancy due to exposure to particulate matter and nitrogen dioxide. Furthermore, approximately 4% of future lung cancer cases and 3% of childhood asthma cases in the immediate area are attributable to the plant's current emissions (Geelen et al., 2023).

Another reason this is relevant has to do with international guidelines. Currently, the Netherlands meets PM2.5 emission standards that are imposed by the European Union (EU). But the value that the World Health Organization (WHO) advises was exceeded in the whole country in 2023 (CLO, 2024b). Furthermore, the EU is set to lower its PM2.5 emission standards from 25 µg/m³ to 10 µg/m³ by the year 2030. Since these future EU standards are not yet met everywhere across the Netherlands, Dutch policy towards reducing PM2.5 emissions might have to become stricter in the upcoming years.

Part 2 - Data Sourcing

2.1 Load in the data

```r
library(readr)
Part1 <- read_csv("data/DataExtract.csv")
```

```
## Rows: 111690 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): Country Or Territory, NUTS Code, NUTS Name, Degree Of Urbanisation...
## dbl (12): Year, Population, Affected Population, Area [km2], Air Pollution A...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
Part2 <- read_csv("data/DataExtract2.csv")
```

```
## Rows: 141255 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): Country Or Territory, NUTS Code, NUTS Name, Degree Of Urbanisation...
## dbl (12): Year, Population, Affected Population, Area [km2], Air Pollution A...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
combined_data <- rbind(Part1,Part2)
opnamen <- read_csv("data/DataExtract2.csv")
```

```
## Rows: 141255 Columns: 24
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (12): Country Or Territory, NUTS Code, NUTS Name, Degree Of Urbanisation...
## dbl (12): Year, Population, Affected Population, Area [km2], Air Pollution A...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

2.2 Provide a short summary of the dataset(s)

```r
head(combined_data)
```

```
## # A tibble: 6 x 24
##   `Country Or Territory` `NUTS Code` `NUTS Name` `Degree Of Urbanisation`   Year
##   <chr>                  <chr>       <chr>       <chr>                     <dbl>
## 1 Netherlands            NL          Netherlands (12) Dispersed Rural Are~  2010
## 2 Netherlands            NL          Netherlands (12) Dispersed Rural Are~  2011
## 3 Netherlands            NL          Netherlands (12) Dispersed Rural Are~  2011
## 4 Netherlands            NL          Netherlands (12) Dispersed Rural Are~  2011
## 5 Netherlands            NL          Netherlands (12) Dispersed Rural Are~  2012
## 6 Netherlands            NL          Netherlands (12) Dispersed Rural Are~  2012
## # i 19 more variables: `Air Pollutant` <chr>, `Data Aggregation Id` <chr>,
## #   Scenario <chr>, Category <chr>, Outcome <chr>, `Health Indicator` <chr>,
## #   Sex <chr>, `Description Of Age Group` <chr>, Population <dbl>,
## #   `Affected Population` <dbl>, `Area [km2]` <dbl>,
## #   `Air Pollution Average [ug/m3]` <dbl>,
## #   `Air Pollution Population Weighted Average [ug/m3]` <dbl>, Value <dbl>,
## #   `Value - lower CI` <dbl>, `Value - upper CI` <dbl>, ...
```

In the combined_data we see 25 variables with 252,945 data points. The dataset is already filtered for PM2.5 pollution (measured in ug/m3) in the Netherlands specifically and it includes data from the year 2005 to 2023. There are three unique Categories; mortality, morbidity and total burden of disease. Outcome has seven unique cases including stroke and lung cancer as well as an eight case listed as “All causes”. Health indicator has 4 categories including Years of life lost and Attributable deaths. Similairly there are four age groups in the dataset. In section 3.1 the data is filtered down to 4806 observations and further used under the variable cleaned_pollution_data.

The Ziekenhuisopnamen dataset shows the hospitalizations across 25 GGD regions in the Netherlands, these are categorized by 48 different diagnosis during the year 2023 specifically. Since there are 48 different diagnosis we chose to focus on one specifically for our spatial visualization.

```
head(opnamen)
```

```
## # A tibble: 6 x 24
##   `Country Or Territory` `NUTS Code` `NUTS Name` `Degree Of Urbanisation`   Year
##   <chr>                  <chr>       <chr>       <chr>                     <dbl>
## 1 Netherlands            NL          Netherlands All Areas (incl.unclassi~  2021
## 2 Netherlands            NL          Netherlands All Areas (incl.unclassi~  2021
## 3 Netherlands            NL          Netherlands All Areas (incl.unclassi~  2021
## 4 Netherlands            NL          Netherlands All Areas (incl.unclassi~  2021
## 5 Netherlands            NL          Netherlands All Areas (incl.unclassi~  2021
## 6 Netherlands            NL          Netherlands All Areas (incl.unclassi~  2021
## # i 19 more variables: `Air Pollutant` <chr>, `Data Aggregation Id` <chr>,
## #   Scenario <chr>, Category <chr>, Outcome <chr>, `Health Indicator` <chr>,
## #   Sex <chr>, `Description Of Age Group` <chr>, Population <dbl>,
## #   `Affected Population` <dbl>, `Area [km2]` <dbl>,
## #   `Air Pollution Average [ug/m3]` <dbl>,
## #   `Air Pollution Population Weighted Average [ug/m3]` <dbl>, Value <dbl>,
## #   `Value - lower CI` <dbl>, `Value - upper CI` <dbl>, ...
```

2.3 Describe the type of variables included

The variables contain health information. In the GGD dataset, the figures “are divided into sex, age, type of admission, region and diagnosis according to the main and subcatagories of the diagnostic classification developed for the Public Health Foresight Study (VTV) of the Dutch National Institute for Public Health and the Environment (RIVM).”(CBS, 2025) The NUTS dataset is “based on gridded data: ambient air concentrations and population density data” (Ortiz et al., 2021)*

*the methodology as described in this 2021 report is that of the one on hand, as per the EEA (2022)

Part 3 - Quantifying

3.1 Data cleaning

```
cleaned_pollution_data<- combined_data
cleaned_pollution_data$code <- cleaned_pollution_data$`NUTS Code`

combined_data$code <- combined_data$`NUTS Code`
require(stringr)
require(tidyverse)
length(cleaned_pollution_data$code)
```

```
## [1] 252945
```

```r
cleaned_pollution_data <- cleaned_pollution_data %>%
  dplyr::filter(str_length(code) == 5
)
cleaned_pollution_data <- cleaned_pollution_data %>%
  filter(
    Category                    == "Mortality",
    Outcome                     == "All causes",
    `Health Indicator`          == "Attributable deaths (AD)",
    Sex                         == "Total",
    `Description Of Age Group` == ">= 30 years of age",
    Scenario                    == "Baseline from WHO 2021 AQG"
  )
```

"str_length(code) == 5": keeps only NUTS 3 regional level records, removing national and provincial aggregates that would distort comparisons. "Category, Outcome, Health Indicator": pins the analysis to one clearly defined metric (air pollution attributable deaths), preventing mixing of incompatible health outcomes. "Sex =="Total" ": avoids triple-counting by excluding the sex-stratified rows that exist alongside the total."Age >= 30": focuses on the population most affected by pollution-related mortality and ensures age group consistency across regions. "Scenario == Baseline WHO 2021": fixes a single modelling assumption across all records.

```r
Ziekenhuisopnamen_diagnose_indeling_VTV_regio_02062026_164323 <- read_delim("data/Ziekenhuisopnamen__dia
     delim = ";", escape_double = FALSE, trim_ws = TRUE,
     skip = 3)
```

```
## New names:
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...2`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...3`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...4`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...5`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...6`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...7`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...8`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...9`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...10`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...11`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...12`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...13`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...14`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...15`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...16`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...17`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...18`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...19`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...20`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...21`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...22`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...23`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...24`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...25`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...26`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...27`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...28`
```

```
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...29`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...30`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...31`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...32`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...33`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...34`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...35`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...36`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...37`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...38`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...39`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...40`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...41`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...42`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...43`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...44`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...45`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...46`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...47`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...48`
## * `Opnamen per 10 000 inwoners` -> `Opnamen per 10 000 inwoners...49`
```

```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 33 Columns: 49
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (49): Onderwerp, Opnamen per 10 000 inwoners...2, Opnamen per 10 000 inw...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

delim = “;”: CSV files sometimes use semicolons as separators instead of commas. escape_double = FALSE: prevents misreading of quotation marks within text fields. trim_ws = TRUE: strips invisible whitespace that would cause merge failures. skip = 3: skips the metadata rows at the top of the file so only actual data is read. ## 3.2 Generate necessary variables

Variable 1

event_data <- cleaned_pollution_data %>% filter(year >= 2015) %>% group_by(year) %>% summarise(avg_pollution = mean(`Air Pollution Population Weighted Average [ug/m3]`, na.rm = TRUE))

Variable 2

cleaned_pollution_data <- cleaned_pollution_data %>% mutate(period = ifelse(year < 2020, “Before 2020”, “2020 onwards”))

cleaned_pollution_data$period < -factor(cleaned_pollution_data$period, levels = c(“Before 2020”, “2020 onwards”))

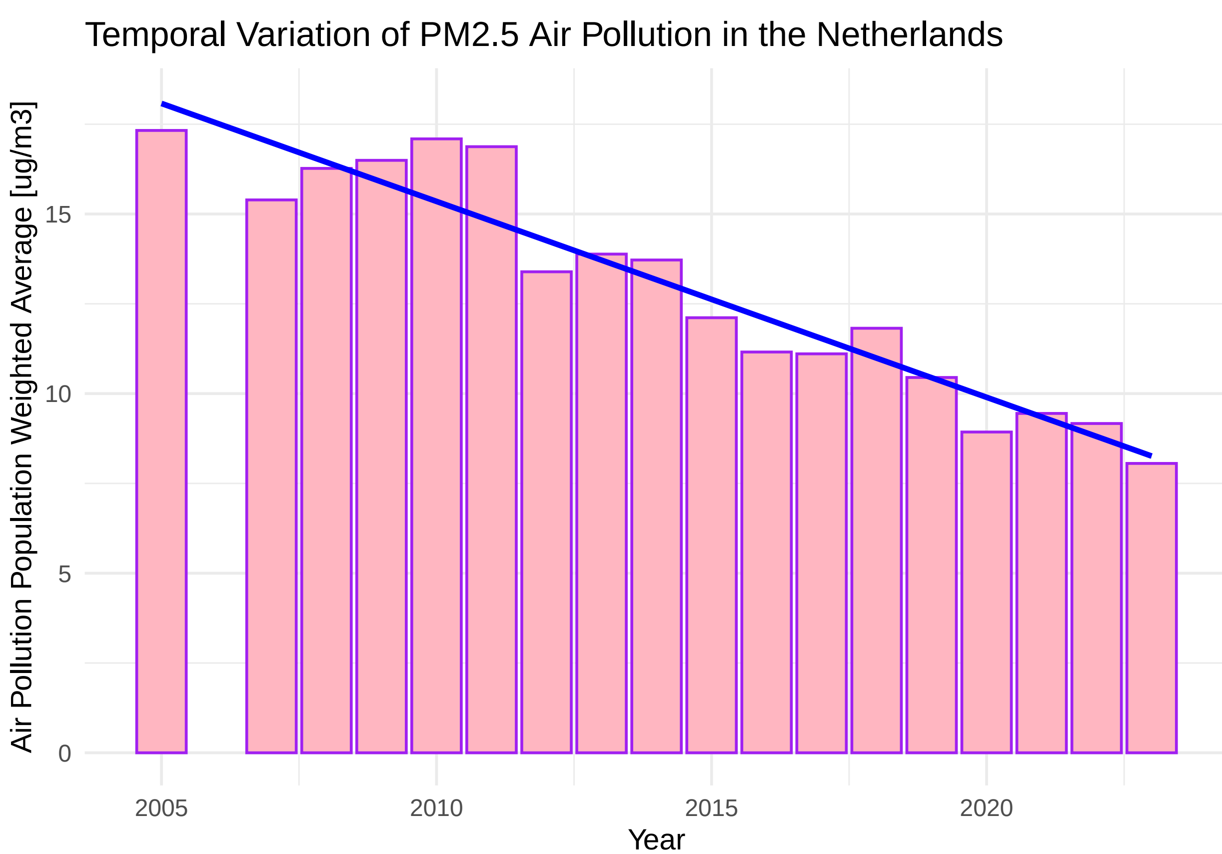
3.3 Visualize temporal variation

```
name=cleaned_pollution_data$year
```

```
## Warning: Unknown or uninitialised column: `year`.
```

```
value=cleaned_pollution_data$`Air Pollution Population Weighted Average [ug/m3]`

require(ggplot2)

ggplot(cleaned_pollution_data, aes(x=Year, y=`Air Pollution Population Weighted Average [ug/m3]`)) +
  geom_bar(stat = "summary", fun = "mean",
           fill = "lightpink",
           colour = "purple") +

  geom_smooth(method = "lm", color = "blue", linewidth = 1) +

  labs(
    title = "Temporal Variation of PM2.5 Air Pollution in the Netherlands",
    y = "Air Pollution Population Weighted Average [ug/m3]",
    x = "Year"
  )+
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The temporal plot is relevant because it shows the longterm exposure to PM2.5 in the Netherlands. This is crucial to understand if air pollution policies have worked.

3.4 Visualize spatial variation

```
library(sf)
```

```
## Linking to GEOS 3.14.1, GDAL 3.12.1, PROJ 9.7.1; sf_use_s2() is TRUE
```

```
library(dplyr)
library(stringr)
library(ggplot2)
library(readr)
library(gridExtra)
```

```
##
## Attaching package: 'gridExtra'
```

```
## The following object is masked from 'package:dplyr':
##
##     combine
```

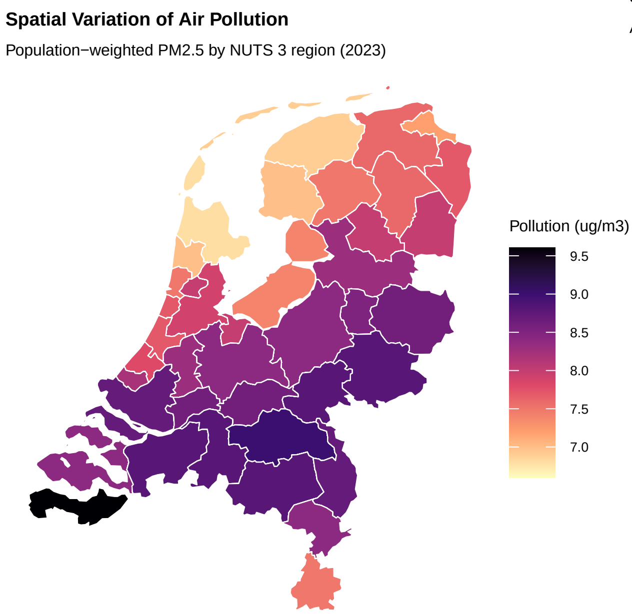
```
nl_map <- st_read("https://cartomap.github.io/nl/wgs84/coropgebied_2023.geojson", quiet = TRUE)

map_data_clean <- cleaned_pollution_data %>%
  filter(Year == 2023)

spatial_joined <- nl_map %>%
  left_join(map_data_clean, by = c("statnaam" = "NUTS Name"))


ggd_map <- st_read("https://cartomap.github.io/nl/wgs84/ggdregio_2023.geojson", quiet = TRUE)

raw_hospital_data <- read_delim("data/Ziekenhuisopnamen__diagnose_indeling_VTV__regio_02062026_164323.c
                                delim = ";", col_names = FALSE, skip = 3)
```

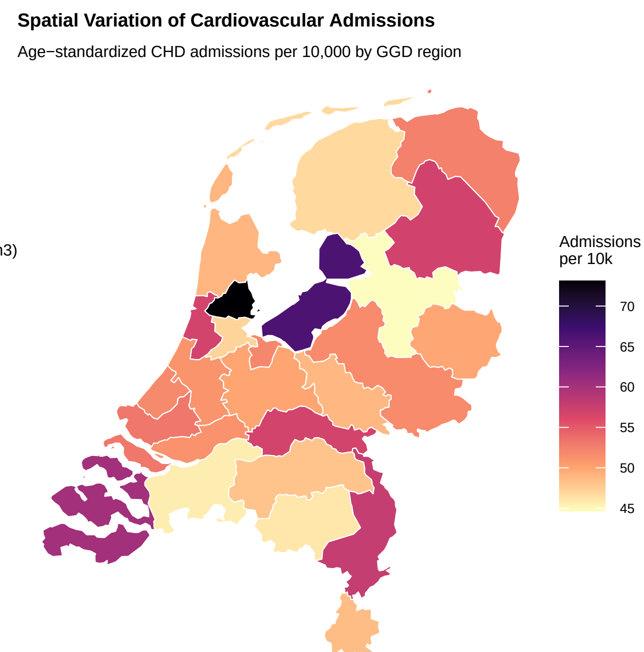
```
## Warning: One or more parsing issues, call `problems()` on your data frame for details,
## e.g.:
##   dat <- vroom(...)
##   problems(dat)
```

```
## Rows: 34 Columns: 49
```

```
## -- Column specification --------------------------------------------------------
## Delimiter: ";"
## chr (49): X1, X2, X3, X4, X5, X6, X7, X8, X9, X10, X11, X12, X13, X14, X15, ...
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
clean_health_data <- data.frame(
  raw_region     = raw_hospital_data$X1[9:33],
  admission_rate = raw_hospital_data$X31[9:33]
) %>%
  mutate(
    admission_rate = as.numeric(str_replace(admission_rate, ",", ".")),
    clean_region   = str_replace(raw_region, " \\(GG\\)", "")
  ) %>%
  mutate(
    clean_region = case_when(
      str_detect(clean_region, "Gelderland-Midden") ~ "Veiligheids- en Gezondheidsregio Gelderland-Midde
      str_detect(clean_region, "ZHZ")               ~ "Dienst Gezondheid & Jeugd ZHZ",
      TRUE ~ clean_region
    )
  )

spatial_joined_ggd <- ggd_map %>%
  left_join(clean_health_data, by = c("statnaam" = "clean_region"))

plot1 <- ggplot(spatial_joined) +
  geom_sf(aes(fill = `Air Pollution Population Weighted Average [ug/m3]`), color = "white", linewidth =
  scale_fill_viridis_c(option = "magma", name = "Pollution (ug/m3)", direction = -1) +
  theme_void() +
  theme(
    text = element_text(size = 6),
    plot.title = element_text(size = 7, face = "bold"),
    plot.subtitle = element_text(size = 6)
  ) +
  labs(
    title = "Spatial Variation of Air Pollution",
    subtitle = "Population-weighted PM2.5 by NUTS 3 region (2023)"
  )

plot2 <- ggplot(spatial_joined_ggd) +
  geom_sf(aes(fill = admission_rate), color = "white", linewidth = 0.2) +
  scale_fill_viridis_c(option = "magma", name = "Admissions\nper 10k", direction = -1) +
  theme_void() +
  theme(
    text = element_text(size = 6),
    plot.title = element_text(size = 7, face = "bold"),
    plot.subtitle = element_text(size = 6)
  ) +
  labs(
    title = "Spatial Variation of Cardiovascular Admissions",
    subtitle = "Age-standardized CHD admissions per 10,000 by GGD region",
  )

grid.arrange(plot1, plot2, ncol = 2)
```



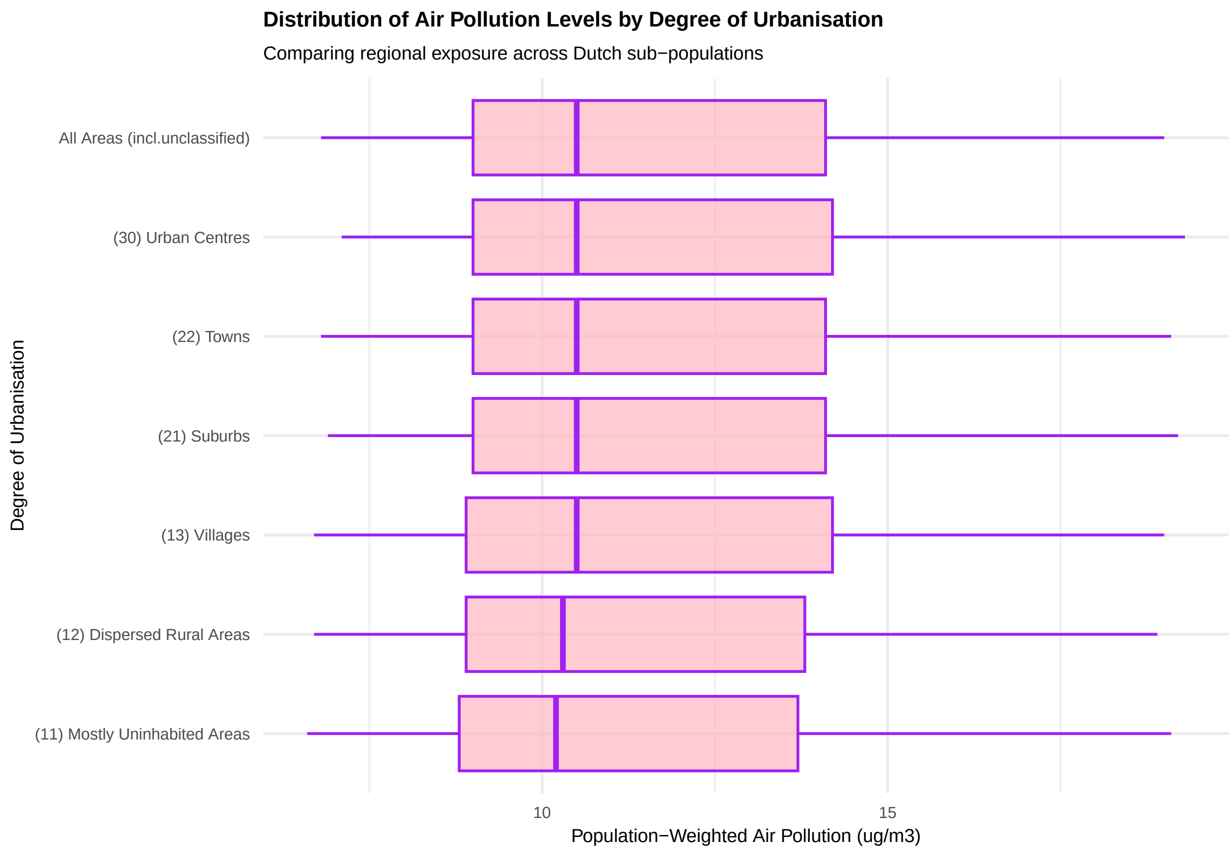


The Spatial maps are relevant because they show regional differences in air pollution and cardiovascular admissions. Because of this policymakers can identify risk areas that need policies.

3.5 Visualize sub-population variation

```r
require(ggplot2)

ggplot(combined_data, aes(x = `Degree Of Urbanisation`, y = `Air Pollution Population Weighted Average

geom_boxplot(
           fill = "lightpink",
           colour = "purple",
           alpha = 0.7) +
  coord_flip() +

  labs(
    title = "Distribution of Air Pollution Levels by Degree of Urbanisation",
    subtitle = "Comparing regional exposure across Dutch sub-populations",
    x = "Degree of Urbanisation",
    y = "Population-Weighted Air Pollution (ug/m3)"
  ) +
  theme_minimal() +
  theme(
    plot.title    = element_text(size = 8, face = "bold"),
```

```r
    plot.subtitle = element_text(size = 7),
    axis.title    = element_text(size = 7),
    axis.text     = element_text(size = 6)
  )
```

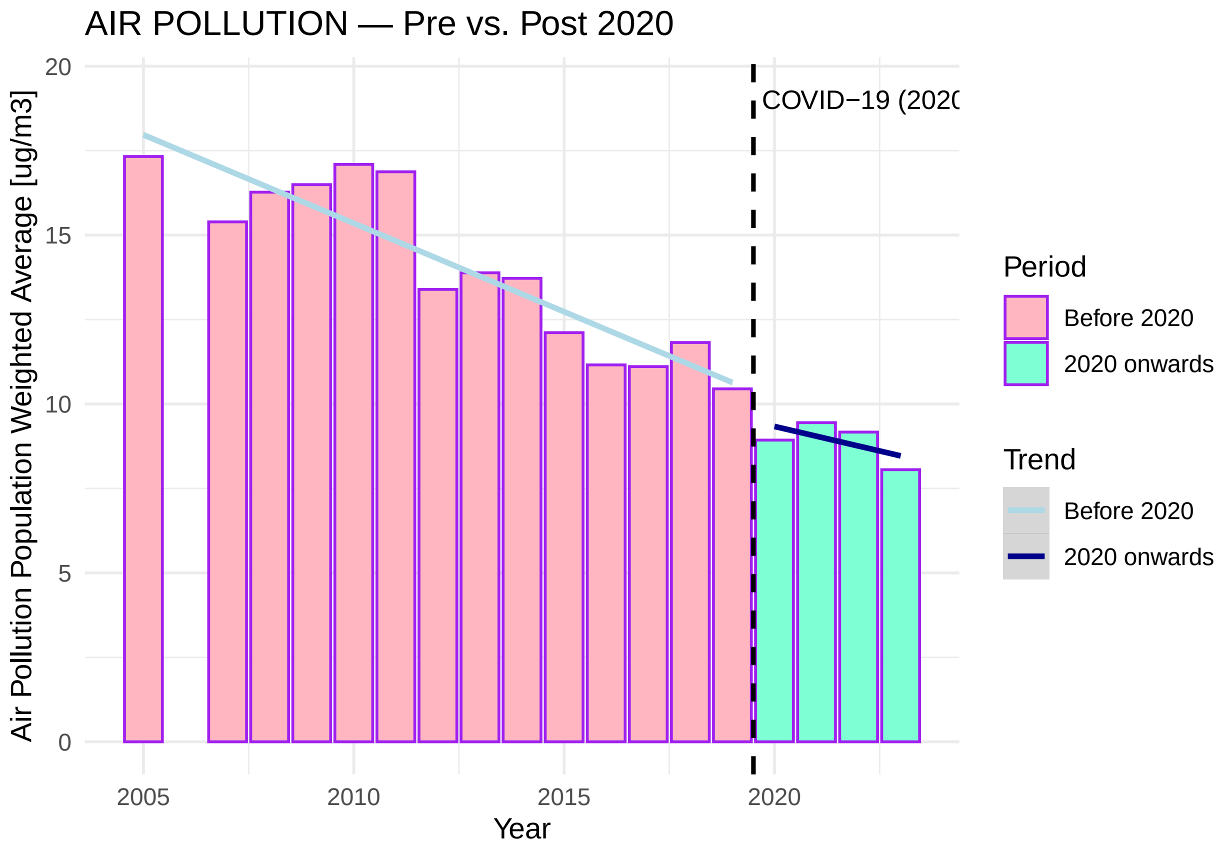


This graph is relevant to our social problem as we are aware that not all regions in The Netherlands stay below the new limits for particulate matter exposure set to be introduced in 2030 by the European Union. With the subpopulation analysis we divide the population based on the degree of urbanisation of the region to highlight certain areas standing out in pollution, which can help assess which regions need more attention towards the new rulesets of 2030.

```r
require(tidyverse)
cleaned_pollution_data <- cleaned_pollution_data %>%
  mutate(period = ifelse(Year < 2020, "Before 2020", "2020 onwards"))

cleaned_pollution_data$period <- factor(
  cleaned_pollution_data$period,
  levels = c("Before 2020", "2020 onwards")
)
ggplot(cleaned_pollution_data,
       aes(x = Year,
           y = `Air Pollution Population Weighted Average [ug/m3]`)) +

  geom_bar(stat = "summary", fun = "mean",
           aes(fill = period),
           colour = "purple") +
```

```r
  geom_smooth(aes(group = period, color = period),
              method = "lm", linewidth = 1, se = TRUE) +

  geom_vline(xintercept = 2019.5,
             linetype = "dashed",
             colour = "black",
             linewidth = 0.8) +

  annotate("text", x = 2019.7,
           y = max(cleaned_pollution_data$`Air Pollution Population Weighted Average [ug/m3]`,
                   na.rm = TRUE),
           label = "COVID-19 (2020)",
           hjust = 0, vjust = 1, size = 3.5, colour = "black") +

  scale_fill_manual(values = c("Before 2020" = "lightpink",
                               "2020 onwards" = "aquamarine")) +
  scale_color_manual(values = c("Before 2020" = "lightblue",
                                "2020 onwards" = "darkblue")) +

  labs(
    title = "AIR POLLUTION - Pre vs. Post 2020",
    y = "Air Pollution Population Weighted Average [ug/m3]",
    x = "Year",
    fill = "Period",
    color = "Trend"
  ) +
  theme_minimal()
```

```
## `geom_smooth()` using formula = 'y ~ x'
```



The covid lockdown analysis is about the change in air pollution before and after the lockdowns of 2020. It could give us a better picture of changes after the event, which could give more depth and future directions in the research of our social problem of tackeling health issues related to air pollution.

Apart from a slightly higher than usual drop in pollution per person in 2020 we can see more meaningful changes when we look at the change in the trend of air pollution per person. Both before and after 2020 the air pollution trends downwards, although the trend before 2020 was a higher decrease than after. There could be a few different reasons why the air pollution per person is decreasing slower post-covid. The population change could have an effect as in the name the pollution is measured per Dutch citizen. Another reason could be that polluting activities increased or a decreased focus on sustainability in The Netherlands. These factors make it hard to make concrete conclusions from this graph alone. Further research into these topics could help further substantiating a meaningful answer.

Part 4 - Discussion

4.1 Discuss your findings

4.2 Analysis of the temporal variation

The temporal variation shows that the population-weighted average air pollution concentration of PM2.5 declined over the period 2005-2023. It falls from approximately 17.3 µg/m³ in 2005 to 8.1 µg/m³ in 2023. A downward trend line is visible with low outliers around 2012 and 2020. There are some fluctuations between the years, but the downward trend is very consistent. This suggests that particulate matter emission policy is effective when it comes to reducing PM2.5 emissions between these years.

4.3 Analysis of the spatial variation

Our spatial analysis shows two maps of the Netherlands. One with the population-weighted PM2.5 by Nuts 3 regions (2023) and one with the age-standardized CHD admissions per 10,000 residents per GGD region (2023.) De Air pollution map shows that the PM2.5 concentrations in 2024 can vary from a bit less than 7,0 μg/m³ tot 9,5 μg/m³. The highest concentrations can be seen Zeeland and in Brabant, the lowest concentrations are in the north of the Netherlands, especially in the north-west of the Netherlands. This seems logical since the highest concentration of residents and intensive industrial businesses operate more in the middle and south of the Netherlands. Though, the relative difference is not that much, it has only a maximum of 2,5 μg/m³ difference in PM2.5. This is also concluded from the boxplot of the subpopulation variation, both show that air pollution is relatively equally spread, probably because of the small scale of the Netherlands. The map of spatial variation of Cardiovascular Admissions shows us that hospital admissions vary stronger between regions. There is a high concentration in Flevoland, which could be explained because of the fact that Flevoland also hospitalizes people from arounding areas like Amsterdam and het Gooi. This means that hospitalization admissions may be an overestimization. It strikes that Zeeland, which has a high Air Pollution level, also has a higher hospitalization number, this is also the case with the Randstad. The regions with higher PM2.5 concentrations also show higher hospitalization numbers of cardiovascular diseases. This supports the hypothesis that long-term exposure to particular matter increases the chance of cardiovascular diseases. But the correlation is not perfectly explainable because of these facts, since other influences like socioeconomic classes or the availability of healthcare for example ar not taken in account here.

4.4 Analysis of the subpopulation variation

The boxplot of the subpopulation variation shows a small and consistent difference in urbanization and air pollution exposure. Urban centers have the highest median population-weighted pollution (~10.6 μg/m³) and the widest spread, while mostly uninhabited and dispersed rural areas sit slightly lower (~10.2–10.4 μg/m³). But the differences between categories are small, this could mean that in the Netherlands, pollution is spatially not very different. This is likely because it is a small country with dense infrastructure and industry. Industrial and travelling emissions are spilled over from dense to sparse areas to some degree. Another explanation for the low variation between levels of urbanization, is that the pollution levels are population-weighted. Areas with a high degree of industrialization and dense infrastructure tend to be more densely populated than regions with less industrial activity, evening out the per capita emissions.

4.5 Event analysis

This graph shows how the air pollution weighted average changes over the years, with a closer look how this changes before and after 2020. This was the year with heavy lockdown measures. Apart from a slightly higher than usual drop in pollution per person in 2020 we can see more meaningful changes when we look at the change in the trend of air pollution per person. Both before and after 2020 the air pollution trends downwards, although the trend before 2020 was a higher decrease than after. There could be a few different reasons why the air pollution per person is decreasing slower post-covid. The population change could have an effect as in the name the pollution is measured per Dutch citizen. Another reason could be that polluting activities increased or a decreased focus on sustainability in The Netherlands. These factors make it hard to make concrete conclusions from this graph alone. Further research into these topics could help further substantiating a meaningful answer.

Part 5 - Reproducibility

5.1 Github repository link

https://github.com/ThijmenGerritsen/Programming-for-Economists-assignment.git

5.2 Reference list

Use APA referencing throughout your document.

1. Janssen, N. et al., (2013). Short-term effects of PM2.5, PM10 and PM2.5–10 on daily mortality in the Netherlands. The Science of the Total Environment, 463–464, 20–26. https://doi.org/10.1016/j.scitotenv.2013.05.062

2. Ultrafijnstof in de lucht. (2024, December 11). Compendium Voor De Leefomgeving. Retrieved June 8, 2026, from https://www.clo.nl/indicatoren/nl062301-ultrafijnstof-in-de-lucht#referentie

3. Geelen, L., Bogers, R., Elberse, J., Houthuijs, D., Montforts, M., & Schuijff, M. (2023). De bijdrage van Tata Steel Nederland aan de gezondheidsrisico's van omwonenden en de kwaliteit van hun leefomgeving. In Rivm (National Institute for Public Health and the Environment). https://doi.org/10.21945/rivm-2023-0171

4. Fijnere fractie van fijn stof (PM2,5) in lucht, 2009-2023. (2024b, July 10). CLO. Retrieved June 8, 2026, from https://www.clo.nl/indicatoren/nl053209-fijnere-fractie-van-fijn-stof-pm25-in-lucht-2009-2023#referentie

5. World Health Organization (WHO). (2006). Air quality guidelines for particulate matter, ozone, nitrogen dioxide and sulfur dioxide. https://www.who.int/publications/i/item/WHO-SDE-PHE-OEH-06-02

6. European Environment Agency. (2022). Air Quality Health Risk Assessments. https://www.eea.europa.eu/en/datahub/datahubitem-view/49930245-dc33-4c47-93b8-9512f0622ebc?activeAccordion=1084339

7. Centraal Bureau voor de Statistiek. (2025). Ziekenhuisopnamen; diagnose-indeling VTV, regio. In Centraal Bureau voor de Statistiek. https://www.cbs.nl/nl-nl/cijfers/detail/84523ned

8. Ortiz, A. G., Gsella, A., Guerreiro, C., Soares, J., Horálek, J., & European Enviroment Agency. (2021). Health risk assessments of air pollution: Estimations of the 2019 HRA, benefit analysis of reaching specific air quality standards and more (ETC/ATNI 2021/10).